

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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Anonymous submission

ABSTRACT

This paper contributes VITA, a zero-shot goal-conditioned value function method that makes frozen Vision–Language Models (VLMs) more useful for robotic progress estimation by adding meta-learned test-time adaptation. The first contribution is an inference-time adaptation mechanism: a lightweight module takes a gradient step on a meta-learned self-supervised loss so that each update is trained to improve value estimation. The second contribution is a temporal-memory mechanism: sequential updates over a trajectory encode history in the adapter parameters, directly targeting the temporal reasoning limitations of frozen VLM representations. The third contribution is a training sampler: dissimilarity-based selection emphasizes semantically diverse trajectory segments and reduces shortcut learning. Across real-world robotic manipulation evaluations, VITA generalizes from a single training environment to out-of-distribution tasks, environments, and embodiments, outperforming the state-of-the-art zero-shot method based on autoregressive VLMs. We further show that VITA value estimates provide effective zero-shot reward shaping for offline reinforcement learning, producing Meta-World multi-task policies that outperform policies trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

The central opportunity behind this paper is to turn the broad visual-semantic knowledge of Vision–Language Models (VLMs) into a reusable learning signal for agents. VLMs generalize across tasks and domains after training on large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). Generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), however, still rely heavily on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) offer a complementary route by producing simulated experience for robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). These trends motivate the contribution we target: a value-learning mechanism that can exploit visual trajectories at scale without requiring new human supervision, even as world models become capable of generating realistic open-ended videos (Hughes et al., 2024; Silver & Sutton, 2025).

One route toward this goal learns policies directly from expert visual trajectories by inferring latent actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). A second route, and the one this paper advances, learns a universal *goal-conditioned value function* that can serve as zero-shot reward shaping for reinforcement learning (RL) and as supervision for imitation learning (Chen et al., 2021; Ma et al., 2022). In this framing, the key prediction problem is reviewer-facing and concrete: given visual observations and a natural language task description, estimate how far the agent has progressed toward completing the task (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Existing VLM value estimators expose two gaps that define our contribution. Contrastive VLM methods (Radford et al., 2021) use similarity between task descriptions and visual observations to ground progress in a shared multimodal space (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024), but frame-level similarity does not encode the temporal context needed to distinguish visually similar states at different task stages, such as folding versus unfolding a shirt. Autoregressive

VLMs (Alayrac et al., 2022) can condition on an entire trajectory, but their progress estimates can inherit monotonic biases from chronologically ordered pre-training data (Ma et al., 2024). More broadly, zero-shot VLM value functions depend on frozen pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023), which limits both out-of-distribution generalization and temporal reasoning (Pătrăucean et al., 2023). Recent alternatives pursue scale, domain-specific fine-tuning, or demonstration-based reward specification (Alakuijala et al., 2024; Ma et al., 2023b; Zhang et al., 2025b; Sontakke et al., 2023); our contribution instead asks whether the representation can adapt at inference time.

We introduce VITA, a zero-shot value function method whose main contribution is to make VLM-based progress estimation adaptive at test time. Rather than relying on new large-scale pre-training or task-specific demonstrations, VITA updates a lightweight adaptation module online so that the value estimator responds to the semantic and temporal context of the current trajectory. At inference, the module takes a gradient step on a meta-learned self-supervised loss (Sun et al., 2024); meta-learning trains this update to improve the downstream value prediction loss (Finn et al., 2017). Because updates are applied sequentially, trajectory history is stored in the adapter parameters (Sun et al., 2024), giving the value function a temporal memory without changing the frozen VLM encoder. A second method contribution is dissimilarity-based training-time sampling, which selects semantically diverse sub-trajectories to reduce shortcut learning (Geirhos et al., 2020). We evaluate these contributions against the state-of-the-art zero-shot setting (Ma et al., 2024) on three reviewer-relevant capabilities: distribution-shift generalization, expert versus non-expert discrimination, and zero-shot reward shaping for offline RL (Levine et al., 2020).

The paper makes the following explicit contributions:

- We propose VITA, a test-time adaptation method that improves the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation without task-specific demonstrations or additional large-scale pre-training.
- We show that VITA generalizes from a single training environment to diverse out-of-distribution tasks, environments, and embodiments in robotic manipulation, outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- We demonstrate that VITA’s zero-shot value estimates can shape rewards for offline RL, yielding Meta-World multi-task learning (MT10) policies that surpass policies trained with the simulator’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

We use goal-conditioned value functions to make the paper’s learning target explicit: task progress in a video is treated as a scalar estimate of goal completion. Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, mapping an observation $o_t \in \mathcal{O}$ and goal specification $g \in \mathcal{G}$ to predicted progress toward the goal. We set $V(o_t; g) = 0$ at task start and $V(o_t; g) = 1$ at completion. Following prior work, expert demonstrations provide supervision by assuming that progress increases monotonically with temporal position (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. This formulation converts unlabeled expert videos into progress targets for learning V , which is the value-estimation contribution used throughout the experiments.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) provides the adaptation principle behind VITA: inference itself becomes a self-supervised learning step (Goodfellow et al., 2016). Instead of requiring labels for a new test trajectory, TTT updates model parameters on the test instance. Although first introduced for static image classification (Sun et al., 2020), sequence-modeling variants (Sun et al., 2024; Wang et al., 2023) update a parametric adaptation module f_{adapt} at each timestep using a gradient step on a self-supervised loss. This is important for our contribution because temporal history is stored

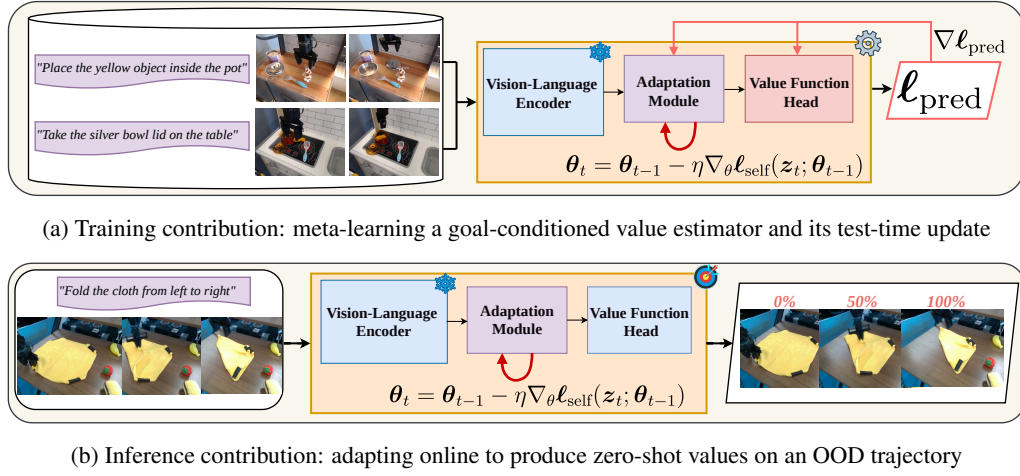


Figure 1: **Overview of VITA.** The method contributes a meta-learned training objective and an inference-time adaptation procedure for zero-shot goal-conditioned value estimation on out-of-distribution trajectories.

in the parameters of f_{adapt} through sequential inference-time updates, rather than only in hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017). The parameters of f_{adapt} therefore act as an implicit memory that preserves temporal order (Sun et al., 2024). Following Sun et al. (2024), we meta-learn the self-supervised task under the gradient-based meta-learning paradigm (Finn et al., 2017): learnable linear projections parameterize the loss, and training optimizes downstream prediction after the update. Our method transfers this idea to goal-conditioned value learning, making meta-learned test-time adaptation a mechanism for better zero-shot progress estimation.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

The architecture is organized around the three contributions needed for zero-shot value estimation. First, a frozen multimodal encoder extracts joint visual-language representations from visual trajectories and task descriptions. Second, a lightweight adaptation module is updated at test time using a self-supervised loss meta-learned specifically to improve value estimation. Third, a regression head maps the adapted representation to a scalar progress value.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We ground VITA in a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), so that the contribution is adaptation rather than additional representation pre-training. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate visual and language features at each timestep t to form joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders. Because CLIP is pre-trained to align paired image-text inputs in a shared representation space (Radford et al., 2021), frames that are semantically closer to goal completion tend to lie closer to the task description; VITA builds on this semantic signal and adds temporal adaptation.

3.1.2 TEST-TIME ADAPTATION

The adaptation module f_{adapt} is the mechanism that turns frozen multimodal features into trajectory-specific value estimates. Following the TTT formulation in Section 2.2, inference updates f_{adapt} online with a gradient step on a meta-learned self-supervised loss ℓ_{self} . The loss is a reconstruction objective built from learnable projections P_K and P_V , and these projections are trained so that performing the

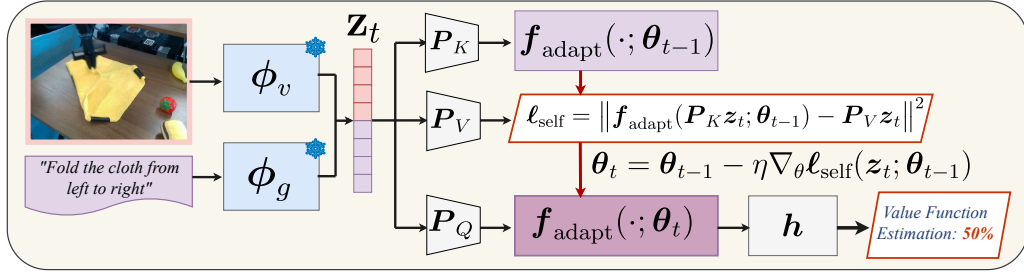


Figure 2: **Test-time Adaptation.** At inference, each timestep t updates f_{adapt} with a meta-learned self-supervised gradient step, making temporal history an explicit parameter-level memory.

self-supervised update improves the supervised value estimate. Specifically, $P_K \in \mathbb{R}^{d' \times 2d}$ creates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ defines the target:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . The key temporal contribution is that these updates are not independent: sequentially updating f_{adapt} lets past observations persist in the parameters and supplies temporal context to later predictions. Section 4.6.2 tests this design directly against trajectory-level and reset-based alternatives.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After adaptation, VITA converts the trajectory-specific representation into a reviewer-interpretable output: a scalar goal-conditioned value. A meta-learned projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps z_t into the adaptation space $\mathbb{R}^{d'}$, and the adapted module f_{adapt} followed by regression head h predicts the value:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), matching value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). Training uses the normalized progress labels y_t defined in Section 2.1, with ℓ_{pred} as the mean squared error between $V(z_t; g)$ and y_t . At inference, h stays frozen while only f_{adapt} is updated by the self-supervised objective. This separation keeps the zero-shot claim precise: no task-specific demonstrations are used at test time, but the value estimator still adapts to the test trajectory.

3.2 TRAINING PROCEDURE

Training has two roles in VITA: it teaches the model what progress means, and it teaches the test-time update how to improve that progress estimate. We therefore optimize a self-supervised loss ℓ_{self} with gradient-based meta-learning so that applying the update reduces the supervised prediction loss ℓ_{pred} (Sun et al., 2024). We train on diverse sub-trajectories chosen by dissimilarity-based sampling, making the sampling strategy part of the contribution rather than an implementation detail: it discourages shortcut learning and emphasizes semantic and temporal evidence.

3.2.1 META-LEARNING

The meta-learning contribution is to train the update rule that will be used at inference. During training, f_{adapt} is updated at each timestep using ℓ_{self} , and we differentiate through those updates with respect to ℓ_{pred} following gradient-based meta-learning (Finn et al., 2017). This jointly optimizes the initialization θ_0 , the projections P_K, P_V, P_Q , and the regression head h . The total training objective combines ℓ_{pred} and ℓ_{self} , weighted by λ , so that θ_0 becomes an initialization from which the

self-supervised test-time step is useful for value estimation. In short, VITA does not merely adapt at test time; it is explicitly trained to adapt for the value-function objective.

3.2.2 DISSIMILARITY-BASED SAMPLING

The sampling contribution targets a specific failure mode of progress learning from videos. Expert visual trajectories contain many redundant neighboring frames (Wang et al., 2018), and a value estimator can exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b), such as frequent late-stage visual patterns, instead of learning semantic progress. We mitigate this by constructing mini-batches from visually diverse sub-trajectories within each trajectory. The resulting batches increase intra-batch variance and act as a diversity-oriented importance sampler that favors semantic coverage.

Formally, for multimodal representations $\{z_1, \dots, z_T\}$, we first create the candidate set $\mathcal{W}$ by extracting fixed-length sub-trajectories with window size w_{tr} and stride s . The desired diverse subset of size k is the one that maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Exact optimization of Eq. 4 would require enumerating all $\binom{|\mathcal{W}|}{k}$ subsets, so we use a scalable heuristic that preserves the intended contribution: each window $w \in \mathcal{W}$ receives a score equal to its total dissimilarity from all other windows:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Selecting the k highest-scoring windows gives a polynomial-time approximation to diverse subset selection. Appendix G shows that the overhead is negligible relative to model training, and Section 4.6.1 validates the contribution empirically by comparing against full-trajectory and random sampling for expert–non-expert discrimination.

4 EXPERIMENTS

The experiments are organized to make the contributions directly testable. We evaluate whether VITA provides: (i) stronger value-estimation generalization under real-world task, environment, and embodiment shifts; (ii) better discrimination between expert and non-expert visual trajectories; and (iii) useful zero-shot reward shaping for offline RL on Meta-World MT10, a simulated multi-task robot-learning benchmark.

4.1 TRAINING SETUP

We deliberately train VITA in a narrow setting so that the generalization contribution is visible. The training data are expert visual trajectories with natural language task descriptions from BridgeData V2 (Walke et al., 2023). We use a curated subset of 2,986 expert pick-and-place demonstrations; it excludes folding, sweeping, and stacking tasks. All training demonstrations use the same robot embodiment (WidowX 250) across four ToyKitchen configurations. We hold out 287 expert demonstrations as the in-distribution test set `tk_pnp`, with examples shown in Appendix A. OpenCLIP ViT-B/32 remains frozen for frame and task-description encoding. The objective combines ℓ_{pred} and ℓ_{self} with $\lambda_{\text{self}} = 0.5$. During training, dissimilarity-based sampling uses window size $w_{\text{tr}} = 8$, $k = 8$ selected sub-trajectories, and stride $s = 1$. At test time, we apply one adaptation step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. Appendix D reports additional implementation details and hyperparameter sweeps.

4.2 BASELINES

The baselines isolate which part of the contribution matters: frozen semantic similarity, supervised progress regression, recurrent temporal modeling, or autoregressive in-context reasoning. VLM-CL estimates values from cosine similarity between CLIP frame and task-description representations (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along

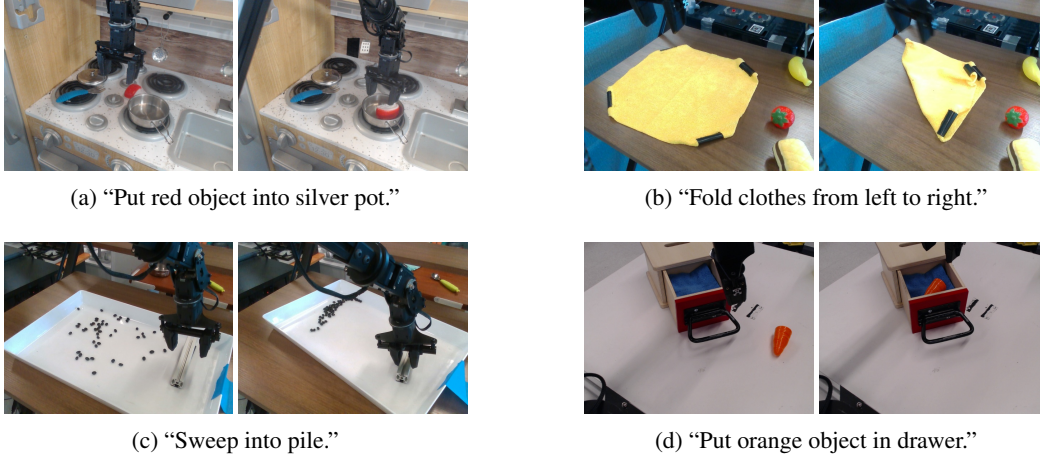


Figure 3: Representative visual trajectories used to test the generalization contribution. (a) In-distribution. (b, c) Environment shift. (d) Embodiment and environment shift.

the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations, and CLIP-GRU adds a Gated Recurrent Unit (GRU) (Cho et al., 2014) over those representations. GVL (Ma et al., 2024) is the state-of-the-art zero-shot autoregressive VLM value function. GVL-0S uses only the test trajectory and task description, while GVL-1S adds one shuffled labeled trajectory from the training distribution as an in-context example. We follow GVL by using Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with the proposed prompt template. We also tested Qwen-VL 2.5 (Yang et al., 2024), which did not overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Appendix D gives implementation details.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: VOC scores testing the generalization contribution under distribution shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
ES	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

This experiment directly tests the generalization contribution by evaluating progress estimators on BridgeData V2 subsets that vary task, environment, and embodiment. Environment shifts change scene layout or background: `lm_pnp` performs pick-and-place in front of a laundry machine; `td_fold`, `ft_fold`, and `rd_fold` perform cloth folding on different surfaces; and `ms_sweep` adds a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training WidowX 250 in morphology and camera perspective. `dt_tk_pnp` and `dt_tk_stack` keep the ToyKitchen environment while changing embodiment, whereas `dt_ft_stack` and `dt_rd_pnp` change both embodiment and environment. Each dataset contains 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 due to

size limitations; Appendix B provides full dataset descriptions. We measure value-estimation quality with *Value Order Correlation* (VOC) (Ma et al., 2024), which evaluates whether predicted progress follows the chronological order of frames. Let $p_1, \dots, p_T$ be predicted values for a trajectory of T frames and let $r_t = t$ be the temporal index; VOC is the Spearman rank correlation ρ_s between predictions and temporal indices.

Table 1 supports the main empirical contribution: methods that preserve temporal information outperform methods that rely only on frozen frame-level semantics or autoregressive prompting. VITA and CLIP-GRU consistently exceed GVL-0S and GVL-1S, while VITA is higher than CLIP-GRU on 6 of 10 evaluation datasets. Both temporal methods outperform non-temporal CLIP baselines (CLIP-FT, VLM-CL, VLM-RM), showing that temporal history is essential for progress estimation. CLIP-FT is competitive with GVL in distribution but does not transfer reliably out of distribution, and VLM-CL and VLM-RM perform poorly without temporal modeling. The GVL variants sometimes benefit from an in-context example and sometimes degrade, with stronger results on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) than on stacking (`dt_tk_stack`, `dt_ft_stack`) or pick-and-place (`dt_tk_pnp`, `lm_pnp`). In contrast, VITA’s more consistent scores across task types and shifts indicate stronger zero-shot generalization than in-context learning in this evaluation.

The long-horizon and embodiment-shift cases clarify where the contribution is most salient. On `ms_sweep`, nearly all methods degrade, but VITA obtains the highest VOC, suggesting that parameter-level temporal memory is more effective than recurrent hidden-state modeling for this long-horizon manipulation task. Under embodiment shift, VITA also remains strong. On `dt_tk_pnp`, where the task and environment match training but the robot changes, both VITA and CLIP-GRU exceed their in-distribution scores, indicating that learned temporal context transfers across embodiments. GVL-1S performs poorly on `dt_tk_pnp` and `dt_tk_stack`, suggesting that the one-shot in-context strategy does not reliably handle embodiment shifts in our setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

The second evaluation asks whether the value estimator contributes more than chronological scoring on expert videos: it should also distinguish expert trajectories from suboptimal ones. We compare predictions on expert and scripted non-expert trajectories collected in the same in-distribution BridgeData V2 setting. The scripted trajectories are generated in ToyKitchen with a heavily randomized controller (Walke et al., 2023) under the same robot embodiment and environment as the expert demonstrations. They cover pick-and-place-like tasks over general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robot`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and a cloth (`sc_pnp_cloth`). Thus, task, embodiment, and environment remain close to training, while the behavior is less smooth and less monotonically progressive than expert behavior. Appendix C shows examples. Because the scripted data are produced by a randomized controller, they do not define calibrated suboptimality levels; nevertheless, a useful value function should assign expert trajectories higher VOC than scripted ones. We report *BinVOC*, $1[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when expert VOC exceeds suboptimal VOC and 0 otherwise.

We average BINVOC over all scripted datasets and compute expert performance VOC_{exp} on the in-distribution `tk_pnp` test set, which contains pick-and-place tasks similar to the scripted evaluation. VITA, GVL-0S, and GVL-1S achieve perfect discrimination by assigning higher VOC to expert trajectories across all scripted tasks. The relevant comparison for our temporal-memory contribution is CLIP-GRU: VITA outperforms it, suggesting that implicit memory from sequential test-time updates is less prone to temporal shortcuts than recurrent hidden states in this setting. Non-temporal CLIP baselines (VLM-CL, VLM-RM) underperform, and CLIP-FT fails on one task, indicating sensitivity to object distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

The third evaluation tests whether VITA’s zero-shot values are useful as rewards, not only as diagnostic progress scores. We use Meta-World MT10 (Yu et al., 2020), which includes ten manipulation tasks such as pick-and-place, door opening, and pushing. VITA is trained only on real-world robotic data and then applied zero-shot in simulation. For each Meta-World task, we generate 20 expert visual demonstrations with the benchmark expert policies and use the goal-conditioned value estimator

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-OS	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Discrimination contribution: expert-versus-scripted success measured by average Bin-VOC across 5 in-distribution scripted datasets. (b) Reward-shaping contribution: Meta-World MT10 offline RL performance measured by interquartile mean (IQM) with 95% stratified bootstrap CIs.

to assign a dense reward to each visual observation. We train multi-task policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021), repeat expert data generation and policy training across 10 random seeds, and evaluate each trained policy on 20 rollouts per task. For each task–seed pair we compute average return, then report interquartile mean (IQM) with 95% stratified bootstrap confidence intervals following Agarwal et al. (2021). Appendix E provides the training and evaluation details.

Table 2 shows that the reward-shaping contribution transfers to downstream policy learning. VITA reaches the highest IQM return (0.815), exceeding all CLIP-based reward baselines. We do not compare to GVL at this scale because it depends on Gemini-1.5, a proprietary autoregressive VLM whose inference cost is prohibitive for RL reward labeling, and open-source alternatives did not provide reliable progress estimates in preliminary tests. We include Meta-World’s fuzzy-logic dense reward (META-WL), which achieves 0.779; VITA exceeds this simulator-provided reward, indicating that a value estimator trained on real-world data can generalize to simulated reward shaping.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

This ablation isolates the sampling contribution. We compare sub-trajectory selection strategies by measuring how well VITA distinguishes expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping windows with stride $s = 1$, which in our setup encourages overfitting to global temporal shortcuts. Random sampling increases diversity by selecting windows uniformly, but it does not explicitly target semantic diversity. Dissimilarity-based sampling selects windows with high pairwise distance in representation space, and it outperforms both alternatives on expert–non-expert discrimination. Appendix F.1 provides the full results.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

This ablation isolates the temporal-memory contribution. VITA adapts step by step without resetting, updating parameters from θ_{t-1} so that trajectory history accumulates directly in the adapter. We compare this implicit-memory design with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which makes one batched gradient update over the full trajectory and therefore averages away temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets to θ_0 at every step and adapts only to the current observation; and (iii) *explicit memory (TTT-EX)*, which resets at every step but updates on a local recent-frame window matching the meta-training window size ($w_{tr} = 8$). Across datasets, VITA outperforms all three, showing that sequential accumulation in parameters is the effective temporal mechanism for value estimation. Appendix F.2 gives the complete analysis.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Prior VLM value estimators clarify what VITA contributes beyond existing zero-shot reward models. Contrastive VLMs have been used as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), but their frame-level similarity scores do not model temporal history. Domain-specific CLIP fine-tuning (Fan et al., 2022; Jiang et al., 2024) can produce dense rewards, but the required supervision weakens zero-shot applicability. RoboCLIP (Sontakke et al., 2023) avoids fine-tuning for one-shot reward specification, yet still requires demonstrations and yields sparse rewards. Large-scale multimodal pre-training methods (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) rely on scale to learn language-conditioned value functions, while ReWiND (Zhang et al., 2025b) uses augmented large-scale data, video rewinding, and a few in-domain demonstrations. VITA’s distinct contribution is to adapt a pre-trained contrastive encoder at inference, adding semantic and temporal context through meta-learned test-time adaptation without new large-scale pre-training or domain-specific demonstrations. Autoregressive VLMs (Alayrac et al., 2022) support in-context trajectory reasoning and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024); GVL (Ma et al., 2024) shuffles frames to mitigate monotonic bias but thereby discards temporal order. VITA preserves chronological order and stores temporal context through sequential parameter updates, which is the key distinction from both frame-level contrastive methods and shuffled autoregressive prompting.

5.2 TEST-TIME ADAPTATION

Test-time adaptation is the methodological basis for our contribution because it offers parameter-efficient adaptation without full fine-tuning. Prompt-based approaches optimize model parameters using test-time task context, although the mechanisms can be difficult to interpret (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Evidence that CLIP encodes human-interpretable concepts (Bricken et al., 2023) has motivated the view that CLIP representations can support sparse test-time adaptation (Lim et al., 2025). Other work optimizes parameters at test time with CLIP-based similarity as a reward signal (Zhao et al., 2024). We build specifically on test-time training (Sun et al., 2020), where an adaptation module is updated with a self-supervised loss at each timestep. Unlike meta-reinforcement learning (Finn et al., 2017; Bauer et al., 2023), which needs task-specific adaptation episodes during training, this paradigm permits direct online adaptation at inference without task labels; VITA contributes a value-function-specific instantiation of that idea.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

The results support three contribution claims. First, meta-learned test-time adaptation improves zero-shot value estimation under task, environment, and embodiment shifts in robotic manipulation. Second, the same value estimator distinguishes expert from non-expert visual trajectories, showing that it captures more than monotonic progress on demonstrations. Third, its zero-shot values are useful enough to shape rewards for offline RL. The mechanism behind these results is sequential parameter-level memory: by updating over a trajectory, VITA encodes semantic and temporal context in the adapter parameters and outperforms both non-temporal CLIP baselines and autoregressive in-context VLM baselines. Its advantage over CLIP-GRU on expert–non-expert discrimination and offline RL further suggests that implicit memory from test-time updates can generalize better than recurrent hidden states for zero-shot value estimation. Pre-trained autoregressive VLMs (GVL) remain powerful general models, but they are not explicitly trained for progress estimation, which requires temporally grounded reasoning over visual trajectories. In our experiments, the runtime overhead of this contribution is small because only the lightweight adaptation module is updated at test time.

6.2 LIMITATIONS AND FUTURE WORK

The contribution also has clear boundaries. Although VITA improves zero-shot progress estimation on unseen tasks and environments, high execution variability and very long horizons may still challenge per-timestep adaptation. The adapter is lightweight, but updating a value estimator at every timestep can raise deployment-safety concerns in real-time control settings. Future work should therefore study alternative VLM adaptation mechanisms, especially for training agents inside world models, and should evaluate VITA in closed-loop control and more complex RL environments. Because the method is agnostic to the underlying vision-language encoder, it could also be applied to domain-specific pre-trained multimodal representations to improve their generalization and temporal reasoning. Finally, our experiments show that dissimilarity-based sampling mitigates shortcut learning, but a theoretical account of when diversity-based sampling prevents shortcuts remains outside this paper’s core contribution.

7 CONCLUSION

VITA contributes a zero-shot value-function framework that makes VLM progress estimation adaptive, temporal, and useful for downstream learning. The method meta-learns a self-supervised test-time update, uses sequential adapter updates as implicit temporal memory, and trains with dissimilarity-based sampling to reduce shortcut learning. Experiments on real-world robotic manipulation show generalization from a single training environment to shifted tasks, environments, and embodiments; expert–non-expert discrimination confirms that the learned values are not merely frame-level semantic scores; and Meta-World results show that the same zero-shot values can shape rewards for offline RL. Together, these results support the paper’s central claim: test-time adaptation is a practical mechanism for improving the generalization and temporal reasoning of VLM-based goal-conditioned value functions.

REFERENCES

- Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, Erik Barker, Tiffany Cai, Prithvijit Chatopadhyay, Yongxin Chen, Yin Cui, Yifan Ding, et al. Cosmos world foundation model platform for physical ai. *arXiv preprint arXiv:2501.03575*, 2025.
- Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, 2021.
- Minttu Alakuijala, Reginald McLean, Isaac Woungang, Nariman Farsad, Samuel Kaski, Pekka Martinen, and Kai Yuan. Video-language critic: Transferable reward functions for language-conditioned robotics. *arXiv preprint arXiv:2405.19988*, 2024.
- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. *arXiv preprint arXiv:2204.14198*, 2022.
- Bowen Baker, Ilge Akkaya, Peter Zhokov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pretraining (vpt): Learning to act by watching unlabeled online videos. In *Advances in Neural Information Processing Systems*, 2022.
- Jakob Bauer, Kate Baumli, Feryal M. P. Behbahani, Avishkar Bhoopchand, Nathalie Bradley-Schmieg, Michael Chang, Natalie Clay, Adrian Collister, Vibhavari Dasagi, Lucy Gonzalez, Karol Gregor, Edward Hughes, Sheleem Kashem, Maria Loks-Thompson, Hannah Openshaw, Jack Parker-Holder, Shreya Pathak, Nicolas Perez Nieves, Nemanja Rakicevic, Tim Rocktäschel, Yannick Schroecker, Satinder Singh, Jakub Sygnowski, Karl Tuyls, Sarah York, Alexander Zacherl, and Lei M. Zhang. Human-timescale adaptation in an open-ended task space. In *International Conference on Machine Learning*, 2023.

- Kate Baumli, Satinder Baveja, Feryal M. P. Behbahani, Harris Chan, Gheorghe Comanici, Sebastian Flennerhag, Maxime Gazeau, Kristian Holsheimer, Dan Horgan, Michael Laskin, Clare Lyle, Hussain Masoom, Kay McKinney, Volodymyr Mnih, Alexander Neitz, Fabio Pardo, Jack Parker-Holder, John Quan, Tim Rocktäschel, Himanshu Sahni, Tom Schaul, Yannick Schroecker, Stephen Spencer, Richie Steigerwald, Luyu Wang, and Lei Zhang. Vision-language models as a source of rewards. *arXiv preprint arXiv:2312.09187*, 2023.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Krzysztof Choromanski, Tianli Ding, Danny Driess, Kumar Avinava Dubey, Chelsea Finn, Peter R. Florence, Chuyuan Fu, Montse Gonzalez Arenas, Keerthana Gopalakrishnan, Kehang Han, Karol Hausman, Alexander Herzog, Jasmine Hsu, Brian Ichter, Alex Irpan, Nikhil J. Joshi, Ryan C. Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Sergey Levine, Henryk Michalewski, Igor Mordatch, Karl Pertsch, Kanishka Rao, Krista Reymann, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Pierre Sermanet, Jaspiar Singh, Anikait Singh, Radu Soricut, Huong Tran, Vincent Vanhoucke, Quan Ho Vuong, Azyaan Wahid, Stefan Welker, Paul Wohlhart, Ted Xiao, Tianhe Yu, and Brianna Zitkovich. Rt-2: Vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning*, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Maria Elisabeth Bechtle, Feryal Behbahani, Stephanie C.Y. Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando De Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *International Conference on Machine Learning*, 2024.
- Annie S Chen, Suraj Nair, and Chelsea Finn. Learning generalizable robotic reward functions from in-the-wild human videos. *arXiv preprint arXiv:2103.16817*, 2021.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using rnn encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*, 2014.
- Nitish Dashora, Dibya Ghosh, and Sergey Levine. Viva: Video-trained value functions for guiding online rl from diverse data. *arXiv preprint arXiv:2503.18210*, 2025.
- Yuqing Du, Ksenia Konyushkova, Misha Denil, Akhil Raju, Jessica Landon, Felix Hill, Nando de Freitas, and Serkan Cabi. Vision-language models as success detectors. In *Conference on Lifelong Learning Agents*, 2023.
- Ashley Edwards, Himanshu Sahni, Yannick Schroecker, and Charles Isbell. Imitating latent policies from observation. In *International Conference on Machine Learning*, 2019.
- Lasse Espeholt, Hubert Soyer, Remi Munos, Karen Simonyan, Vlad Mnih, Tom Ward, Yotam Doron, Vlad Firoiu, Tim Harley, Iain Dunning, Shane Legg, and Koray Kavukcuoglu. IMPALA: Scalable distributed deep-RL with importance weighted actor-learner architectures. In *International Conference on Machine Learning*, 2018.
- Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu, and Anima Anandkumar. Minedojo: Building open-ended embodied agents with internet-scale knowledge. In *Advances in Neural Information Processing Systems*, 2022.
- Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning*, 2017.

-
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2020.
- Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep learning*. MIT Press, 2016.
- David Ha and Jürgen Schmidhuber. World models. In *Advances in Neural Information Processing Systems*, 2018.
- Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering atari with discrete world models. In *International Conference on Learning Representations*, 2021.
- Jonathan Ho and Stefano Ermon. Generative adversarial imitation learning. In *Advances in Neural Information Processing Systems*, 2016.
- Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural computation*, 1997.
- Anthony Hu, Lloyd Russell, Hudson Yeo, Zak Murez, George Fedoseev, Alex Kendall, Jamie Shotton, and Gianluca Corrado. GAIA-1: A generative world model for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- Edward Hughes, Michael D. Dennis, Jack Parker-Holder, Feryal M. P. Behbahani, Aditi Mavalankar, Yuge Shi, Tom Schaul, and Tim Rocktäschel. Position: Open-endedness is essential for artificial superhuman intelligence. In *International Conference on Machine Learning*, 2024.
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- Haobin Jiang, Junpeng Yue, Hao Luo, Ziluo Ding, and Zongqing Lu. Reinforcement learning friendly vision-language model for minecraft. In *European Conference on Computer Vision*, 2024.
- Muhammad Uzair Khattak, Hanoona Rasheed, Muhammad Maaz, Salman Khan, and Fahad Shahbaz Khan. Maple: Multi-modal prompt learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. Openvla: An open-source vision-language-action model. In *Conference on Robot Learning*, 2024.
- Ilya Kostrikov, Ashvin Nair, and Sergey Levine. Offline reinforcement learning with implicit q-learning. *arXiv preprint arXiv:2110.06169*, 2021.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021a.
- Youngwoon Lee, Andrew Szot, Shao-Hua Sun, and Joseph J. Lim. Generalizable imitation learning from observation via inferring goal proximity. In *Advances in Neural Information Processing Systems*, 2021b.
- Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643*, 2020.
- Jianxiong Li, Jinliang Zheng, Yinan Zheng, Liyuan Mao, Xiao Hu, Si Cheng, Haoyi Niu, Jihao Liu, Yu Liu, Jingjing Liu, Ya-Qin Zhang, and Xianyu Zhan. Decisionnnc: Embodied multimodal representations via implicit preference learning. In *International Conference on Machine Learning*, 2024.

-
- Hyesu Lim, Jinho Choi, Jaegul Choo, and Steffen Schneider. Sparse autoencoders reveal selective remapping of visual concepts during adaptation. In *International Conference on Learning Representations*, 2025.
- Ziyi Lin, Honglie Liu, Andrew Zisserman, and Andrea Vedaldi. Frozen clip models are efficient video learners. In *European Conference on Computer Vision*, 2022.
- Xudong Ma, Chen Liang, Zhecan Wang, Chuang Gan, and Zhiting Li. Swapprompt: Test-time prompt adaptation for vision-language models. In *Advances in Neural Information Processing Systems*, 2023a.
- Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training. *arXiv preprint arXiv:2210.00030*, 2022.
- Yecheng Jason Ma, Vikash Kumar, Amy Zhang, Osbert Bastani, and Dinesh Jayaraman. LIV: Language-image representations and rewards for robotic control. In *International Conference on Machine Learning*, 2023b.
- Yecheng Jason Ma, Joey Hejna, Ayzaan Wahid, Chuyuan Fu, Dhruv Shah, Jacky Liang, Zhuo Xu, Sean Kirmani, Peng Xu, Danny Driess, Ted Xiao, Jonathan Tompson, Osbert Bastani, Dinesh Jayaraman, Wenhao Yu, Tingnan Zhang, Dorsa Sadigh, and Fei Xia. Vision language models are in-context value learners. In *International Conference on Learning Representations*, 2024.
- Yutaka Matsuo, Yann LeCun, Maneesh Sahani, Doina Precup, David Silver, Masashi Sugiyama, Eiji Uchibe, and Jun Morimoto. Deep learning, reinforcement learning, and world models. *Neural Networks*, 2022.
- Viorica Pătrăucean, Yusuf Aytar, Roozbeh Mottaghi, Oleh Rybkin, Mario Lucic, Carl Vondrick, William T. Freeman, and Andrew Zisserman. Perception test: A diagnostic benchmark for multimodal video models. In *Advances in Neural Information Processing Systems*, 2023.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.
- Scott Reed, Konrad Zolna, Emilio Parisotto, Sergio Gomez Colmenarejo, Alexander Novikov, Gabriel Barth-Maron, Mai Giménez, Yury Sulsky, Jackie Kay, Jost Tobias Springenberg, Tom Eccles, Jake Bruce, Ali Razavi, Ashley D. Edwards, Nicolas Manfred Otto Heess, Yutian Chen, Raia Hadsell, Oriol Vinyals, Mahyar Bordbar, and Nando de Freitas. A generalist agent. *arXiv preprint arXiv:2205.06175*, 2022.
- Juan Rocamonde, Victoriano Montesinos, Elvis Nava, Ethan Perez, and David Lindner. Vision-language models are zero-shot reward models for reinforcement learning. In *International Conference on Learning Representations*, 2024.
- Dominik Schmidt and Minqi Jiang. Learning to act without actions. In *International Conference on Learning Representations*, 2024.
- David Silver and Richard S. Sutton. Welcome to the era of experience. Technical report, Google AI, 2025.
- SIMA Team, Maria Abi Raad, Arun Ahuja, Catarina Barros, Frederic Besse, Andrew Bolt, Adrian Bolton, Bethanie Brownfield, Gavin Buttimore, Max Cant, et al. Scaling instructable agents across many simulated worlds. *arXiv preprint arXiv:2404.10179*, 2024.
- Sumedh Sontakke, Jesse Zhang, Sébastien M. R. Arnold, Karl Pertsch, Erdem Biyik, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. Roboclip: One demonstration is enough to learn robot policies. In *Advances in Neural Information Processing Systems*, 2023.
- Yu Sun, Xiaolong Wang, Zhuang Liu, John Miller, Alexei A. Efros, and Moritz Hardt. Test-time training with self-supervision for generalization under distribution shifts. In *International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2020.

-
- Yu Sun, Xinhao Li, Karan Dalal, Jiarui Xu, Arjun Vikram, Genghan Zhang, Yann Dubois, Xinlei Chen, Xiaolong Wang, Sanmi Koyejo, Tatsunori Hashimoto, and Carlos Guestrin. Learning to (learn at test time): Rnns with expressive hidden states. *arXiv preprint arXiv:2407.04620*, 2024.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.
- Homer Rich Walke, Kevin Black, Tony Z. Zhao, Quan Vuong, Chongyi Zheng, Philippe Hansen-Estruch, Andre Wang He, Vivek Myers, Moo Jin Kim, Max Du, Abraham Lee, Kuan Fang, Chelsea Finn, and Sergey Levine. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning*, 2023.
- Limin Wang, Yuanjun Xiong, Dahua Lin, and Luc Van Gool. Temporal segment networks: Towards good practices for deep action recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2018.
- Renhao Wang, Yu Sun, Yossi Gandelsman, Xinlei Chen, Alexei A. Efros, and Xiaolong Wang. Test-time training on video streams. *arXiv preprint arXiv:2307.05014*, 2023.
- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Ze Yang, Yiming Zhang, Jerry Yu, Junru Cai, and Ming-Hsuan Yang. UniSim: A neural closed-loop sensor simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on Robot Learning*, 2020.
- Chuheng Zhang, Tim Pearce, Pushi Zhang, Kaixin Wang, Xiaoyu Chen, Wei Shen, Li Zhao, and Jiang Bian. What do latent action models actually learn? *arXiv preprint arXiv:2506.15691*, 2025a.
- Jiahui Zhang, Yusen Luo, Abrar Anwar, Sumedh Anand Sontakke, Joseph J Lim, Jesse Thomason, Erdem Biyik, and Jesse Zhang. RewiND: Language-guided rewards teach robot policies without new demonstrations. In *Conference on Robot Learning*, 2025b.
- Shuyuan Zhao, Xin Wang, Linchao Zhu, and Yi Yang. Test-time adaptation with clip reward for zero-shot generalization in vision-language models. *arXiv preprint arXiv:2401.12345*, 2024.
- Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 2022.

A TRAINING DATASET

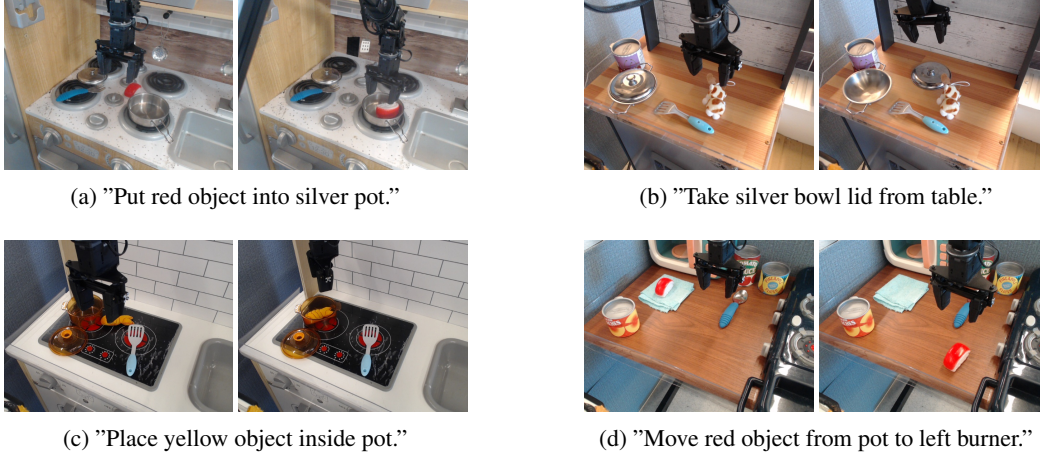


Figure 4: Training examples used to learn the value-estimation and adaptation objectives. Each subfigure shows start and end frames from an expert demonstration with its natural language task description, collected across four ToyKitchen environments.

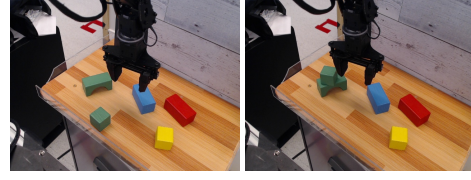
B EVALUATION DATASETS

Table 3: Evaluation datasets grouped by the contribution they test: task, environment, and embodiment generalization.

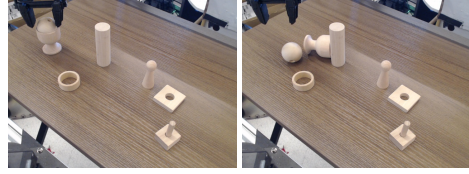
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



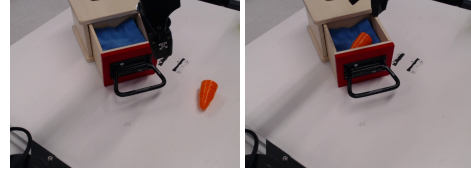
(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."



(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

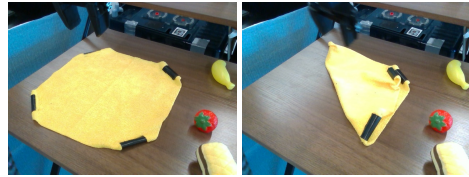
Figure 5: Embodiment-shift examples used to test zero-shot generalization. Each subfigure shows start and end frames with the task description; the top row changes only the robot (DeepThought in ToyKitchen), while the bottom row also changes the environment.



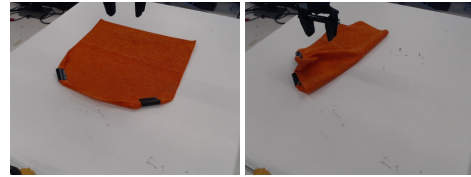
(a) lm_pnp: "Place blue cloth inside washer."



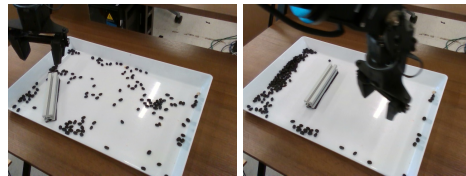
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Environment-shift examples used to test whether VITA transfers beyond the training scene. Each subfigure shows start and end frames with the task description, using the same robot embodiment across visually and structurally different environments.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Scripted non-expert examples used for the discrimination contribution. Each subfigure shows start and end frames from a ToyKitchen trajectory with the WidowX 250 robot; tasks involve pick-and-place behavior over varied object categories, with generic descriptions because public annotations are unavailable.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

The implementation keeps the contribution focused on adaptation rather than backbone training. We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, encoding each visual observation and task description with the CLIP vision and text encoders. We use a concatenated joint representation rather than an element-wise product because it improves validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. Training uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup. We use batch size 32 and pad trajectories to 120 frames, matching the longest training sequence. The self-supervised loss weight λ_{self} is 0.5, selected by validation performance. We train for 5 epochs because validation VOC plateaus early and longer training does not improve results. Unless otherwise specified, dissimilarity-based sampling uses stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows. All experiments were run on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference, only the temporal adaptation module f_{adapt} is updated, using the same projection dimension $d' = 64$ as in training. The default setting applies one gradient step ($t_{\text{ep}} = 1$) with learning rate $\eta = 0.1$. Table 4 reports the sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$. The sweep shows that a single larger step ($\eta = 0.1$, $t_{\text{ep}} = 1$) and multiple

Table 4: Validation sweep for the test-time adaptation contribution: VOC as a function of learning rate η and adaptation steps t_{ep} .

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

smaller steps ($\eta = 0.01$, $t_{\text{ep}} = 10$) yield comparable validation VOC. We use the single-step setting because it preserves the adaptation contribution while minimizing inference cost.

D.3 BASELINES

The appendix baselines are chosen to isolate VITA’s adaptation and temporal-memory contributions. VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected as described above. CLIP-FT keeps the frozen CLIP encoder and supervised regression head but removes the adaptation module and self-supervised loss; it uses a linear projection and two-layer MLP trained with standard supervised regression, without meta-learning or test-time adaptation. To increase expressivity, CLIP-FT uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt describing the environment. CLIP-GRU keeps VITA’s frozen CLIP encoder and regression head h but replaces f_{adapt} with a one-layer GRU of hidden size 64, whose final hidden state is passed to h . For both GVL-0S and GVL-1S, we use `gemini-1.5-pro-latest` (Gemini Team et al., 2024); the original GVL implementation used `gemini-1.5-pro`. We also evaluated `Qwen-VL 2.5` (Yang et al., 2024), which did not overcome temporal bias despite frame shuffling, and GPT-4o (Hurst et al., 2024), which consistently declined to produce scalar progress estimates. All GVL evaluations use the original prompt template (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

For the reward-shaping contribution, we keep the offline RL learner fixed across all reward sources and use Implicit Q-Learning (IQL). Expectile regression uses $\tau = 0.7$, advantage weighting temperature 3.0, and clipping threshold 100.0. The actor and critic learning rates are 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies train for 100,000 gradient steps and are evaluated every 10,000 steps on 20 rollouts per task with horizon 150. The implementation follows `d3rlpy`, with configuration: `expectile=0.7`, `weight_temp=3.0`, `max_weight=100.0`, `actor_lr=1e-4`, `critic_lr=3e-4`, `batch_size=256`.

E.2 EVALUATION PROTOCOL

To make the downstream reward-shaping comparison robust, we follow Agarwal et al. (Agarwal et al., 2021) and report the interquartile mean (IQM) of returns across all task–seed pairs. IQM averages the middle 50% of scores, reducing sensitivity to outliers relative to the mean or median. We estimate 95% confidence intervals with 10,000 stratified bootstrap resamples, stratified by task, and apply the same protocol to all Meta-World MT10 methods.

Table 5: Sampling-ablation evidence for the dissimilarity contribution. Disc@VOC across scripted datasets marks whether a strategy successfully discriminates expert from suboptimal trajectories. **FT**, **Rand**, and **Diss** denote full-trajectory, random, and dissimilarity-based sampling parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

This appendix expands the sampling ablation and shows why dissimilarity-based selection is part of the method contribution. We evaluate how training-time sub-trajectory sampling affects expert-versus-suboptimal discrimination. Table 5 reports Disc@VOC over 5 scripted datasets, indicating whether a model assigns higher average VOC to expert `tk_pnp` trajectories than to each scripted dataset. The comparison includes: (1) full-trajectory sampling, which adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following (Sun et al., 2024); (2) random sampling, which uniformly selects k sub-trajectories of length w_{tr} per video; and (3) dissimilarity-based sampling, which builds a candidate set with stride s and selects k windows maximizing pairwise distance in representation space.

The results make the sampling contribution visible: dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ achieves perfect discrimination. Using fewer windows with the same length ($k = 4$, $w_{tr} = 8$) matches this performance, while shortening windows to $w_{tr} = 4$ reduces effectiveness. Full-trajectory sampling performs worst, supporting the concern that adapting over all consecutive windows can overfit to global temporal shortcuts. Random sampling improves over full-trajectory sampling but is less reliable than dissimilarity-based selection. These findings show that both semantic diversity and local temporal structure matter for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Temporal-memory ablation: validation VOC scores under distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We compare four adaptation strategies to isolate the temporal-memory mechanism: (1) TTT-TR applies one offline gradient update over the full trajectory, producing a batched loss that averages across frames and removes temporal order; (2) TTT-RS resets the adaptation module at every step and updates only on the current observation, making it memoryless; (3) TTT-EX also resets at every step but updates on a local recent-frame window with the meta-training size $w_{\text{tr}} = 8$, so gradients are averaged within the window and ordering is discarded; (4) VITA (OURS) updates incrementally at every timestep, forming implicit memory through accumulated adapter parameters. We use the same test-time hyperparameter sweep for all variants, as described in Appendix D. VITA, TTT-RS, and TTT-TR perform best with learning rate 0.1 and one adaptation step, while TTT-EX performs best with learning rate 1.0 and one step.

Table 6 shows that TTT-TR, TTT-RS, and TTT-EX have similar performance, meaning that batched adaptation over either the full trajectory or a short explicit window behaves much like the memoryless reset baseline. VITA consistently outperforms all three, making sequential per-timestep updates the salient temporal contribution. For implicit memory, local-window updates would double-count recent observations because they appear both explicitly in the window and implicitly in accumulated parameters. Updating less frequently would also discard intermediate frames. Thus, the test-time training mechanism that works in this setting is sequential updating without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

This analysis shows that the dissimilarity-sampling contribution is computationally practical. Let $\mathcal{W}$ be the candidate set with $N_c = |\mathcal{W}|$. Instead of enumerating all $\binom{N_c}{b}$ subsets from Eq. 4, we score each candidate window by the sum of pairwise distances to all other windows, i.e., the row sum of the distance matrix. With $N_c = T - w_{\text{tr}} + 1$, each flattened window has dimension $m = w_{\text{tr}} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU with batched `cdist`, requiring $\mathcal{O}(N_c^2 \cdot w_{\text{tr}} \cdot d)$ operations.

In our experiments, trajectories are short enough that this diversity objective is inexpensive: the mean length is 69 and the maximum length is 120. With batch size 32, feature dimension $d = 1024$ for the CLIP-based multimodal representation, and window length $w_{\text{tr}} = 8$, the worst-case cost is:

$$32 \cdot N_c^2 \cdot w_{\text{tr}} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This cost is negligible relative to a single model forward/backward pass, while enabling the training batches to emphasize diverse windows.